

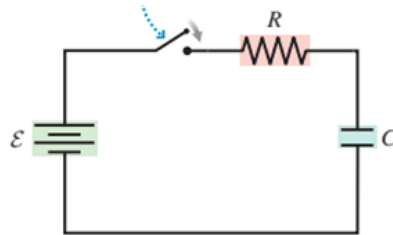
RC Circuits

ENGS124 Electricity and Magnetism Lab

18 November 2025

Theory

When we connect a charged capacitor and resistor, the capacitor starts to discharge. However, depending on the resistance, the current changes.



After applying Kirchoff's law, the following equation is reached:

$$IR = \frac{Q}{C}$$

The current is denoted the direction of positive charges, and the current is slowly decreasing, as the capacitor discharges. Therefore, we get $I = -\frac{dQ}{dt}$:

$$-\frac{dQ}{dt}R = \frac{Q}{C}$$

$$\frac{dQ}{Q} = -\frac{dt}{RC}$$

$$\ln(Q) = -\frac{t}{RC} + c$$

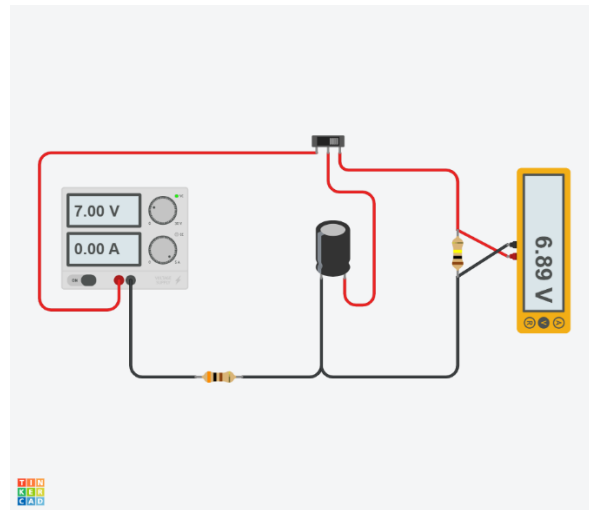
$$Q = Q_0 e^{-\frac{t}{RC}}, V = V_0 e^{-\frac{t}{RC}}, I = I_0 e^{-\frac{t}{RC}}$$

The experiment is carried out to test this equation and investigate the capacitance of different connection combinations of capacitors.

Part 1 Single Capacitor

The following circuit was assembled, with $R_1 = 325.8 \Omega$ and $R_2 = 98.9 k\Omega$. The first resistor was chosen to limit the charging current, and the second value was chosen to slow down the capacitor discharge enough to make the experiment both accurate and not take an eternity. The voltage was taken a constant 7 V. The capacitor was first charged, then the discharge on the second resistor was measured, during set intervals.

Setup



Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Capacitor	1	Component that stores charges
Resistors	2	For limiting the current and discharge
Connection Wires	~	Copper wires
Breadboard	1	A platform for assembling the circuit
Switch	1	Two position switches with three legs
Stopwatch	1	To measure time

Variables

Name	Type	Resolution
Time	Measured	0.01 s

V_1	Measure	0.01 V
V_{all}	Control	0.01 V
R_1 and R_2	Parameter	0.01 Ω

Errors

Name	Type	Impact
Voltmeter Internal Resistance	Systematic	Voltmeter itself has some internal resistance and while connected in parallel to the resistor it might affect the readings.
Capacitor Leakage	Systematic	The capacitor degrades and overtime it can lose its charge, even without a connected load.

Formula Used to Calculate the Current in the Circuit

$$V(t) = V_0 e^{-\frac{t}{RC}}$$

$$\frac{V(t)}{V_0} = e^{-\frac{t}{RC}}$$

$$\ln\left(\frac{V_R}{V_{all}}\right) = -\frac{t}{RC}$$

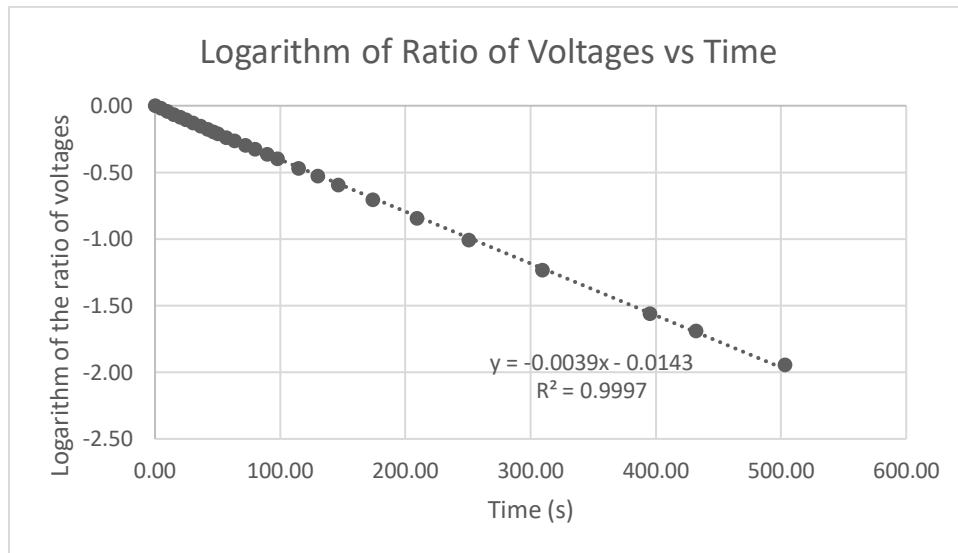
$$C = -\frac{t}{R \ln\left(\frac{V_R}{V_{all}}\right)}$$

Data

V_{all} (V)	R_2 (Ω)	V_R (V)	t (s)	$\ln \frac{V_R}{V_{all}}$	C (μ F)
7.00	98900.00	7.00	0.00	0.00	~
7.00	98900.00	6.88	4.85	-0.02	2836.04
7.00	98900.00	6.70	9.95	-0.04	2296.82
7.00	98900.00	6.56	15.01	-0.06	2337.81
7.00	98900.00	6.41	20.10	-0.09	2308.16
7.00	98900.00	6.29	24.78	-0.11	2342.76
7.00	98900.00	6.15	30.30	-0.13	2366.56

7.00	98900.00	6.00	36.30	-0.15	2381.03
7.00	98900.00	5.86	42.27	-0.18	2404.37
7.00	98900.00	5.75	46.85	-0.20	2408.16
7.00	98900.00	5.68	50.41	-0.21	2439.27
7.00	98900.00	5.51	57.16	-0.24	2414.74
7.00	98900.00	5.37	63.61	-0.27	2426.32
7.00	98900.00	5.19	72.48	-0.30	2449.60
7.00	98900.00	5.04	79.88	-0.33	2458.67
7.00	98900.00	4.85	89.44	-0.37	2464.62
7.00	98900.00	4.69	97.86	-0.40	2470.76
7.00	98900.00	4.38	114.84	-0.47	2476.58
7.00	98900.00	4.12	130.18	-0.53	2483.28
7.00	98900.00	3.86	146.54	-0.60	2489.23
7.00	98900.00	3.46	173.76	-0.70	2493.36
7.00	98900.00	3.01	209.44	-0.84	2509.21
7.00	98900.00	2.56	250.87	-1.01	2521.72
7.00	98900.00	2.04	309.30	-1.23	2536.50
7.00	98900.00	1.47	395.29	-1.56	2561.03
7.00	98900.00	1.29	432.26	-1.69	2584.26
7.00	98900.00	1.00	503.06	-1.95	2613.97

Plot

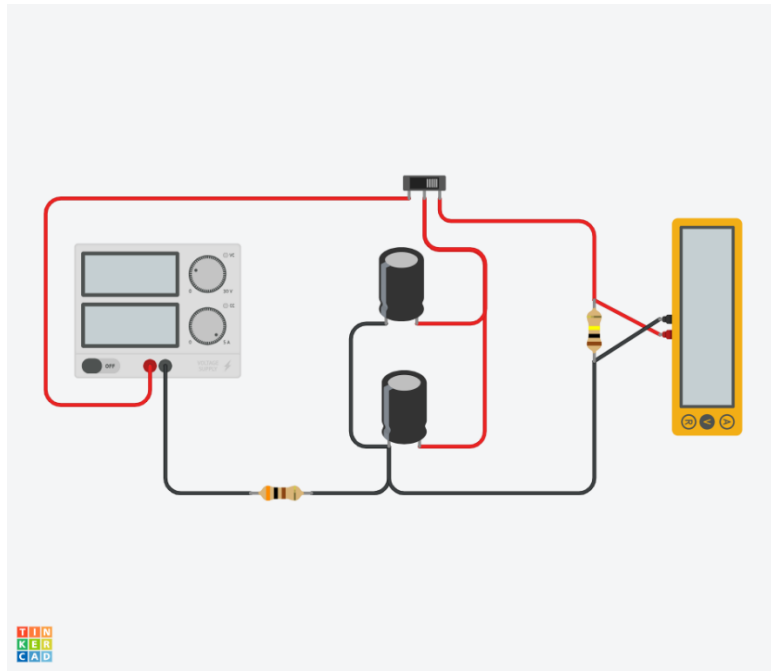


As we can see above the relationship between time and voltage is confirmed and the average capacitance is calculated to be $C = 2364.42 \mu\text{F}$. Which has an error percentage of 7.45% from the real value of $C = 2200 \mu\text{F}$. Therefore, it is reasonable to assume that the theoretical equation is valid.

Part 2 Light Dependent Resistor (LDR)

The same experiment was carried out, but the one capacitor was swapped with two capacitors in parallel. All the error sources, tools, materials, and experiment related procedures and protocols remain the same.

Setup

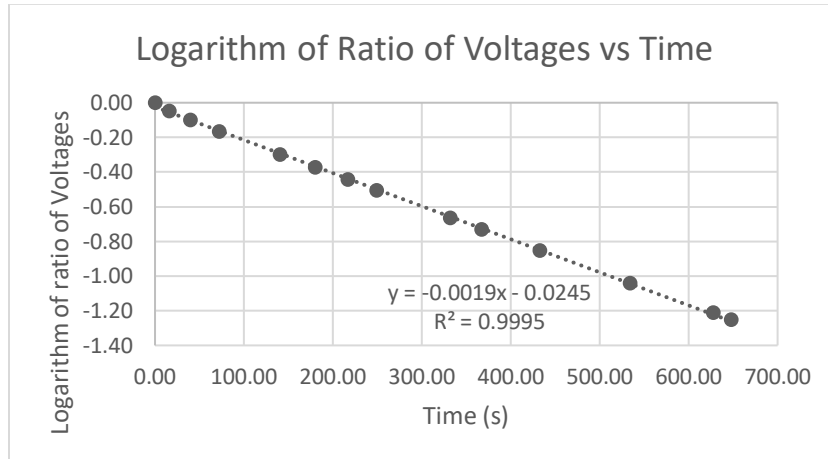


Data

V_{all} (V)	R_2 (Ω)	V_R (V)	t (s)	$\ln \frac{V_R}{V_{all}}$	C (μ F)
7.00	98,900.00	7.00	0.00	0.00	~
7.00	98,900.00	6.67	16.13	-0.05	3377.37
7.00	98,900.00	6.33	39.73	-0.10	3992.84
7.00	98,900.00	5.92	72.14	-0.17	4352.85
7.00	98,900.00	5.19	140.38	-0.30	4744.40
7.00	98,900.00	4.81	179.98	-0.38	4850.09
7.00	98,900.00	4.48	216.89	-0.45	4913.93
7.00	98,900.00	4.22	249.25	-0.51	4979.94
7.00	98,900.00	3.60	332.33	-0.66	5053.21
7.00	98,900.00	3.37	367.20	-0.73	5079.14
7.00	98,900.00	2.98	432.84	-0.85	5124.84
7.00	98,900.00	2.47	534.29	-1.04	5186.11

7.00	98,900.00	2.08	627.89	-1.21	5231.57
7.00	98,900.00	2.00	648.07	-1.25	5230.66

Plot

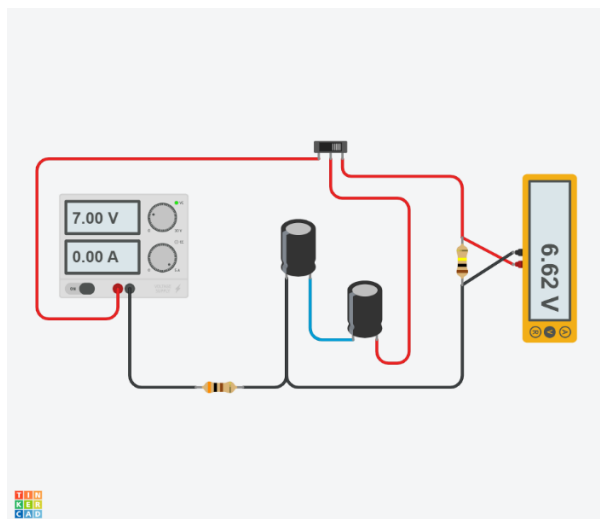


From the graph it is apparent that the relation between time and voltage is confirmed, and the average capacitance is calculated to be $C = 4678.23 \mu\text{F}$, and the real value of $C_1 + C_2 = 4400 \mu\text{F}$, their similarity has an error of 6.32%. The formula is valid.

Part 3 Two Series Capacitors

The same experiment was carried out, but the one capacitor was replaced by two capacitors connected in series. All the error sources, tools, materials and experiment related procedures and protocols remain the same.

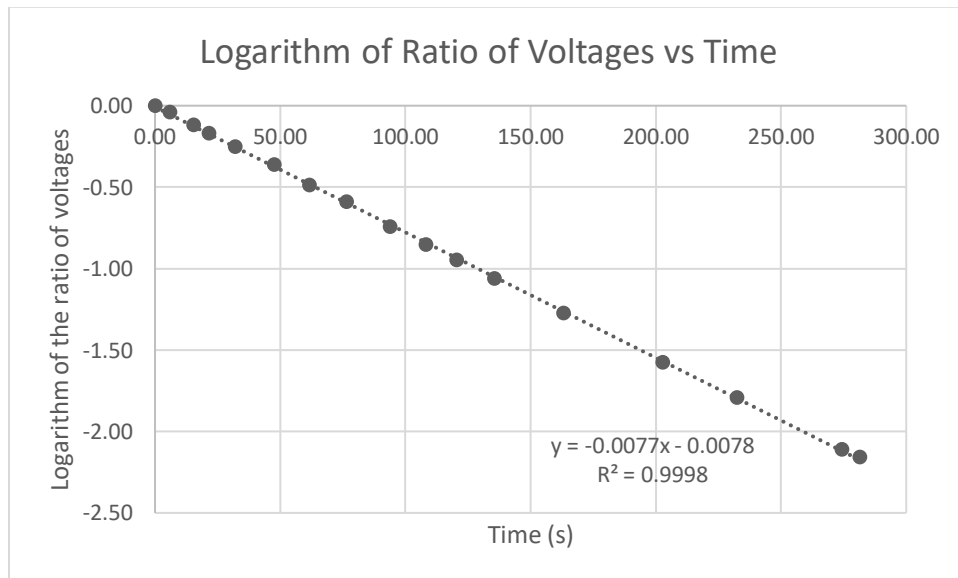
Setup



Data

V_{all} (V)	R_2 (Ω)	V_R (V)	t (s)	$\ln \frac{V_R}{V_{all}}$	C (μ F)
7.02	98900.00	7.02	0.00	0.00	~
7.02	98900.00	6.73	5.90	-0.04	1414.05
7.02	98900.00	6.22	15.41	-0.12	1287.79
7.02	98900.00	5.92	21.45	-0.17	1272.60
7.02	98900.00	5.44	32.06	-0.25	1271.32
7.02	98900.00	4.88	47.68	-0.36	1325.85
7.02	98900.00	4.30	61.73	-0.49	1273.42
7.02	98900.00	3.89	76.57	-0.59	1311.44
7.02	98900.00	3.34	93.97	-0.74	1279.16
7.02	98900.00	2.99	108.25	-0.85	1282.43
7.02	98900.00	2.72	120.45	-0.95	1284.52
7.02	98900.00	2.43	135.46	-1.06	1291.08
7.02	98900.00	1.96	163.20	-1.28	1293.41
7.02	98900.00	1.45	202.70	-1.58	1299.48
7.02	98900.00	1.17	232.43	-1.79	1311.64
7.02	98900.00	0.85	274.44	-2.11	1314.33
7.02	98900.00	0.81	281.56	-2.16	1318.33

Plot



The calculated capacitance is $C = 1301.93 \mu\text{F}$, and $\frac{C_1 C_2}{C_1 + C_2} = 1100 \mu\text{F}$, they are similar within an error of 18.36 %.